# Twenty Years of Time Series Econometrics in Ten Pictures

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# Table of Contents

1. Introduction
2. SVAR
3. DFM
4. Model Instability and Latent Variables
5. Forecasting
6. DSGE Estimation
7. Inference
8. Conclusion

1 Introduction

2 SVAR

3 DFM

4 Model Instability and Latent Variables

5 Forecasting

6 DSGE Estimation

7 Inference

8 Conclusion

## A Shock

A shock is an unanticipated, autonomous change linked to a specific source. The main objective can be described as: **after observing a policy change, how can we determine its causal effect?**

Much progress has been made toward answering this question, and the authors describe six areas:

- estimation of dynamic causal effects;
- estimation of dynamic structural models with optimizing agents (DSGE!);
- methods for exploiting information in "big data" that are specialized to economic time series;
- improved methods for forecasting and for monitoring the economy;
- tools for modeling time variation in economic relationships; and
- improved methods for statistical inference.

*The schemes used to identify structural shocks in vector autoregressions were often seen as unconvincing by researchers outside the field, and the small structural models of the time were not econometrically estimated, miring that enterprise in an unhelpful debate over how to calibrate such models. In addition, there were chinks emerging in the theoretical econometric underpinnings of inference in time series data, as well as opportunities for using the much larger datasets becoming available, if only the tools to do so could be developed. The time was ripe for progress.*

1 Introduction

2 SVAR

3 DFM

4 Model Instability and Latent Variables

5 Forecasting

6 DSGE Estimation

7 Inference

8 Conclusion

## Definition

We can define the VAR(p) as follows, where $y_t$ is a vector of variables, e.g. $y_t = (\pi_t, i_t)'$:

$$y_t = c + B_1 y_{t-1} + B_2 y_{t-2} + \cdots + B_p y_{t-p} + u_t \tag{1}$$

In the case above, there is no contemporaneous relation between $\pi_t$ and $i_t$. Let us allow for it:

$$A_0 y_t = c + A_1 y_{t-1} + A_2 y_{t-2} + \cdots + A_p y_{t-p} + \epsilon_t \tag{2}$$

## Definition

$$A_0 y_t = c + A_1 y_{t-1} + A_2 y_{t-2} + \cdots + A_p y_{t-p} + \epsilon_t \tag{3}$$

From this VAR(p), we can write:

$$u_t = A_0^{-1} \epsilon_t = B \epsilon_t \tag{4}$$

$u_t$ is the vector of residuals. Suppose there are two shocks: monetary ($\epsilon_t^m$) and supply ($\epsilon_t^s$). The variance-covariance matrix of $u_t$ is $BB'$. So:

$$BB' = \Sigma_u = \begin{pmatrix} \text{Var}(u_t^\pi) & \text{Cov}(u_t^\pi, u_t^i) \\ \text{Cov}(u_t^\pi, u_t^i) & \text{Var}(u_t^i) \end{pmatrix} \tag{5}$$

## An Example

Suppose that:

$$BB' = \Sigma_u = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1.25 \end{pmatrix} \tag{6}$$

But we know that: $\begin{pmatrix} u_t^{\pi} \\ u_t^{i} \end{pmatrix} = u_t = B\epsilon_t = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} \begin{pmatrix} \epsilon_t^{s} \\ \epsilon_t^{m} \end{pmatrix}$

Using the moments of the data, we can recover the parameters of $B$ by solving a system of equations.

## An Example (cont.)

$$\begin{cases} \text{Var}(u_t^\pi) = b_{11}^2 + b_{12}^2 = 1 \\ \text{Var}(u_t^i) = b_{21}^2 + b_{22}^2 = 1.25 \\ \text{Cov}(u_t^\pi, u_t^i) = b_{11}b_{21} + b_{12}b_{22} = 0.5 \end{cases}$$

We have 3 equations and 4 unknowns. **That is the identification problem**. Now let us look at alternatives that add information to this system.

## First Alternative: Cholesky

Based on theory, we can assume that a monetary shock does not affect inflation contemporaneously. Looking back at $B$, this means $b_{12} = 0$, which is the additional equation. Then:

$$B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0.5 & 1 \end{pmatrix} \tag{7}$$

But what if we assume instead that $b_{21} = 0$? Then:

$$B_{alt} = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} = \begin{pmatrix} 0.894 & 0.447 \\ 0 & 1.118 \end{pmatrix} \tag{8}$$

We have identified the system, but now we face another problem: choosing between explanations, since $BB' = B_{alt}B'_{alt} = \Sigma_u$.

## A Short Break: An Alternative View

Let us decompose $B$ as $B(\theta) = PQ(\theta)$, where:

$$Q(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \tag{9}$$

Then: $BB' = PQ(\theta)Q(\theta)'P' = PP' = \Sigma_u$.

In the previous example, $P = \begin{pmatrix} 1 & 0 \\ 0.5 & 1 \end{pmatrix}$

So, $B(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ 0.5\cos\theta + \sin\theta & \cos\theta - 0.5\sin\theta \end{pmatrix}$

The identification problem reduces to finding $\theta$.

## Second Alternative: Sign Restrictions

We can assume that a monetary shock raises the interest rate and does not raise inflation. In the notation above, this is equivalent to:

$$\cos\theta - 0.5\sin\theta > 0 \quad \text{and} \quad -\sin\theta < 0$$

Solving this system, we conclude that $\theta \in (0, \arctan 2) \approx (0°, 63.4°)$, which is set identification.

Problems reported in the paper: inference, and how to narrow the set.

## Third Alternative: External Instruments (Proxy SVAR)

Let $z_t$ be an instrument. We require: $E[z_t \epsilon_t^m] = \alpha > 0$ and $E[z_t \epsilon_t^s] = 0$. This implies:

$$E[u_t z_t] = E[B\epsilon_t z_t] = B\,E[\epsilon_t z_t] = B(0, \alpha)' \tag{10}$$

This instrument provides another equation to identify the system. Solving it for the same example, we find a relation between $b_{12}$ and $b_{22}$, which is equivalent to $\theta = \pi/6$: point identification. One problem with this strategy is weak instruments.

## Fourth Alternative: Heteroskedasticity

Suppose the data show that, in crisis periods, the variance of the monetary shock is four times larger than in normal periods. This gives us more equations to identify the system.

The original one:

$$\Sigma_{u,1} = BB' = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1.25 \end{pmatrix} \tag{11}$$

The second one:

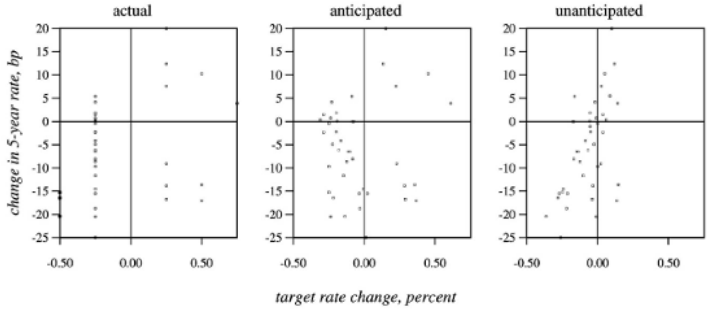
$$\Sigma_{u,2} = B \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix} B' = \begin{pmatrix} 1.75 & -0.42 \\ -0.42 & 2.39 \end{pmatrix} \tag{12}$$

# Kutner (2001)

*Figure 1*

**Changes in the 5-year Treasury Rate and in the Target Federal Funds Rate on Federal Reserve Open Market Committee (FOMC) Announcement Dates**

*(fed funds changes are decomposed into anticipated and unanticipated components using changes in the fed funds futures market on announcement dates)*



1 Introduction

2 SVAR

3 DFM

4 Model Instability and Latent Variables

5 Forecasting

6 DSGE Estimation

7 Inference

8 Conclusion

## DFM

DFMs are an important solution. They address the dimensionality problem of VARs and rest on the hypothesis that, if a large number of series are highly correlated, there is a common factor behind them that explains a large part of their variance.

$$\begin{aligned} X_{it} &= \lambda_i f_t + e_{it} \\ f_t &= \phi f_{t-1} + \mu_t \end{aligned} \tag{13}$$

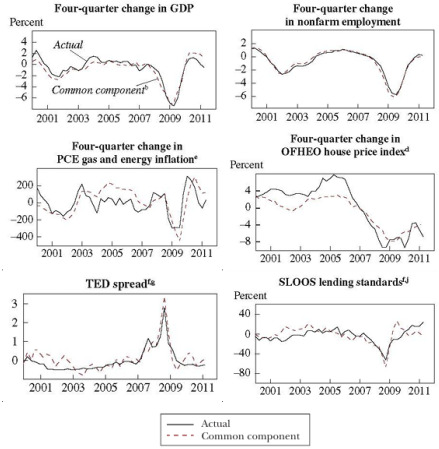
They can be estimated by **principal components** or with the **Kalman filter**, but identification problems remain.

# Stock and Watson (2012)

*Figure 3*

**Selected US macroeconomic Time Series: Actual Values and Common Components**

*(where the common components are the fitted values using the factors from a 200-variable, 6-factor dynamic factor model fit using data from 1959–2007)*



1 Introduction

2 SVAR

3 DFM

4 Model Instability and Latent Variables

5 Forecasting

6 DSGE Estimation

7 Inference

8 Conclusion

## Motivation

Time series relations are **unstable**: e.g. the Great Moderation (lower volatility after 1984), followed by the financial crisis.

- First answer: tests for a single, deterministic **break**. But if there was one break, why not another?
- Alternative: let parameters **evolve stochastically**, i.e., treat them as **latent variables**.

A latent variable exists but is not observed. Example: trend inflation $\tau_t$, hidden inside observed inflation $\pi_t$.

## State-Space Models and the Kalman Filter

Two equations:

$$\pi_t = \tau_t + \varepsilon_t \quad \text{(measurement)}$$
$$\tau_t = \tau_{t-1} + \eta_t \quad \text{(transition)}$$

The **Kalman filter** estimates the state each period:

1. Predict: $\tau_{t|t-1} = \tau_{t-1}$
2. Update: $\tau_t = \tau_{t|t-1} + K_t(\pi_t - \tau_{t|t-1})$

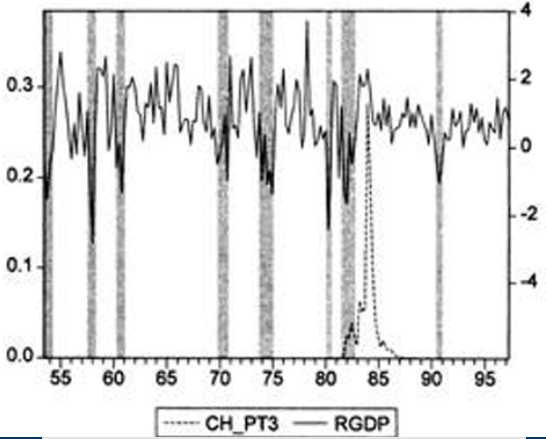
The gain $K_t$ is small if the noise $\varepsilon_t$ is large, and large if the trend shocks $\eta_t$ are large.

If the model is **linear and Gaussian**, the Kalman filter also gives the exact likelihood.

# Kim and Nelson (1999)

*Figure 6*

**US GDP Growth and the Posterior Probability of a Regime Change in its Innovation Variance**



## When the Kalman Filter Is Not Enough

If time-varying parameters multiply **latent** variables, the model becomes nonlinear.
Example: **stochastic volatility**

$$\eta_t = \sigma_t \xi_t, \qquad \ln \sigma_t^2 = \ln \sigma_{t-1}^2 + \nu_t$$

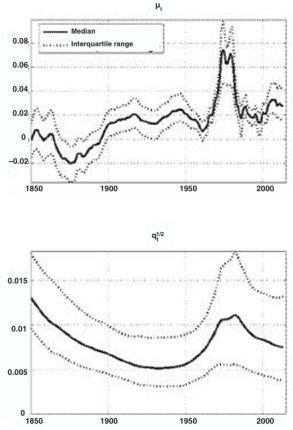
Solution: **MCMC** (Gibbs sampler), alternating simple blocks:

1. draw the trend $\tau_{1:T}$ given the volatilities (Kalman filter);
2. draw the volatilities given the trend;
3. draw the fixed parameters given both;
4. repeat many times $\Rightarrow$ posterior distribution of the latent variables.

# Cogley and Sargent (2015)

*Figure 7*

**Trend Inflation (Upper Panel) and the Standard Deviation of the Trend Innovation (Lower Panel) in an Unobserved Components–Stochastic Volatility Model of US Inflation, 1850–2012**



1 Introduction

2 SVAR

3 DFM

4 Model Instability and Latent Variables

5 Forecasting

6 DSGE Estimation

7 Inference

8 Conclusion

## Introduction

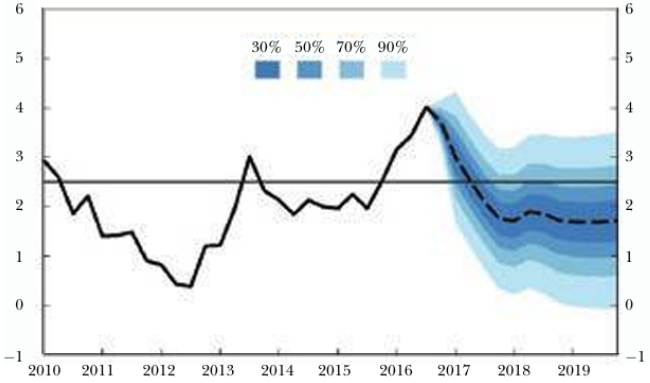
- A huge problem in forecasting is the uncertainty, which hasn't "dramatic" improvements in accuracy over the past decades.
- The profession developed ways to inform it: the fan charts, an alternative to communicate how evolves the distributions of the variables.
- The main agenda on this area is how to forecast better by developing reliable models and tolls for evaluating those models.
- Eliot and Timmermann is an important reference on this topic.

# Norges Bank (2016)

*Figure 4*

**Fan Chart (Density Forecast) for Consumer Price Index (CPI) inflation in Norway**

*(percent; four-quarter change)*

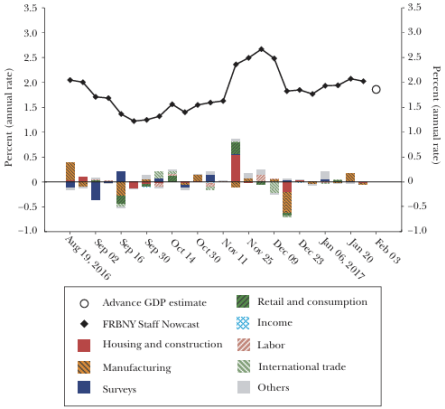


# NY Fed (2017)

*Figure 5*

**Contributions of Daily Data Releases to the Federal Reserve Bank of New York Real-Time Nowcast of 2016Q4 GDP Growth**

*(bars represent weekly contributions of data revisions to changes in the nowcast)*



1 Introduction

2 SVAR

3 DFM

4 Model Instability and Latent Variables

5 Forecasting

6 DSGE Estimation

7 Inference

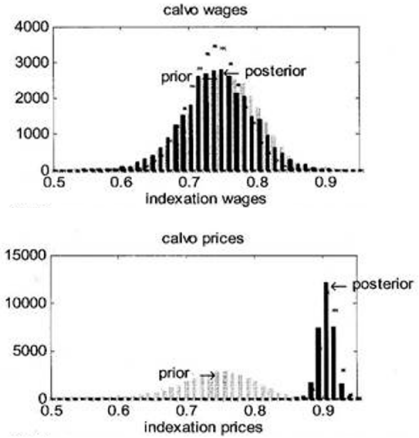
8 Conclusion

## Frame Title

- There's been a lot of advances in DSGE Estimation. The most simple strategies depends on log-linearizated model, in which the prior distribution helps the model estimation, which is by Kalman Filter and MCMC.
- The challenge nowadays remains in non-linear models.

# Smets-Wouters (2003)

*Figure 2*
**Prior and Posterior Distributions for Two Structural Parameters in a Dynamic Stochastic General Equilibrium Model**



1 Introduction

2 SVAR

3 DFM

4 Model Instability and Latent Variables

5 Forecasting

6 DSGE Estimation

7 Inference

8 Conclusion

## Motivation

A 95% confidence interval should contain the true value 95% of the time, and a 5% test should reject a true null 5% of the time.

In the 1990s, studies showed that, with time series data, nominal 5% tests often rejected far more often. Three sources:

- weak instruments (weak identification);
- serially correlated errors;
- highly persistent regressors.

## Weak Instruments

The IV estimator is a ratio:

$$\hat{\beta}_{IV} = \frac{\text{Cov}(y,z)}{\text{Cov}(x,z)}$$

If $\text{Cov}(x,z) \approx 0$, the estimator is biased toward OLS and its distribution is far from normal.

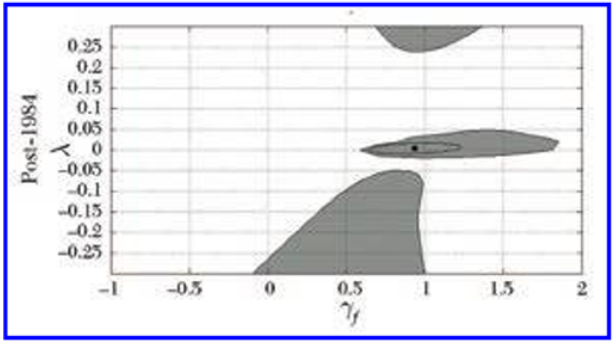
| Instrument | First-stage $F$ | Rejection of a 5% test |
|---|---|---|
| Strong | 52 | 5.5% |
| Weak | 3 | 10.0% |
| Very weak | 1.2 | 13.1% |

- Rule of thumb: worry if $F < 10$; use weak-instrument-robust sets.
- "there are now methods for dealing with weak instruments with very good size and power, both asymptotically and in finite samples [textbook regression with single endogenous regress and errors homoskedastic and serially uncorrelated]. As one departs from this model, most notably when the number of parameters gets large and/or the model is nonlinear in the parameters, the toolkit is less complete

# Mavroeidis, Plagborg-Møller, and Stock (2014)

*Figure 8*

**Point Estimate and 90% Confidence Sets for Hybrid New Keynesian Phillips Curve Parameters: Standard Generalized Method of Moments (Ellipse) and Weak-Instrument Robust (Gray)**



## Serially Correlated Errors (HAR Inference)

With serial correlation, each observation brings less new information. Newey–West standard errors add autocovariances up to a small bandwidth.

| Persistence $\rho$ | Rejection of a 5% Newey–West test |
|---|---|
| 0 | 5.5% |
| 0.5 | 8.1% |
| 0.9 | 27.3% |

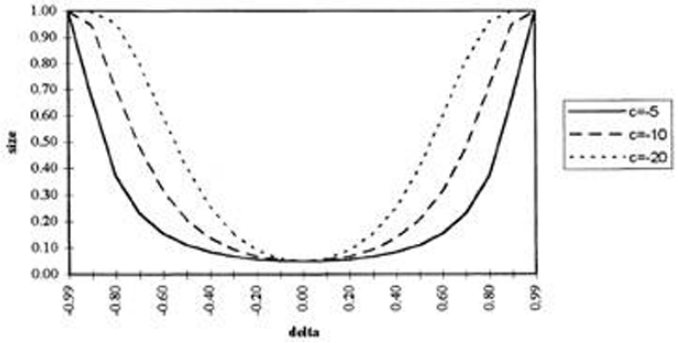
Solution: much larger bandwidths with nonstandard (fixed-$b$) critical values (Kiefer, Vogelsang and Bunzel, 2000).

# Elliot (1998)

*Figure 9*

**Asymptotic Size of Tests of Values of the Cointegrating Coefficient Using Efficient Cointegrating Estimators and Their Standard Errors when the Time Series Follow Local-to-Unity Processes with Parameter *c***

*(delta is the correlation between innovations in the error and in the regressor)*



## Persistent Regressors and Cointegration

Many series have a largest root close to, but not exactly, 1:

$$\rho = 1 + \frac{c}{T}, \qquad \text{e.g. } T = 200,\ c = -5 \Rightarrow \rho = 0.975$$

- A root of 0.975 cannot be distinguished from 1 in typical samples.
- Figure 9 (Elliott, 1998): standard cointegration tests then reject a true null far more than 5%, especially when the correlation $\delta$ between shocks is large.
- There is still no consensus replacement.

1 Introduction

2 SVAR

3 DFM

4 Model Instability and Latent Variables

5 Forecasting

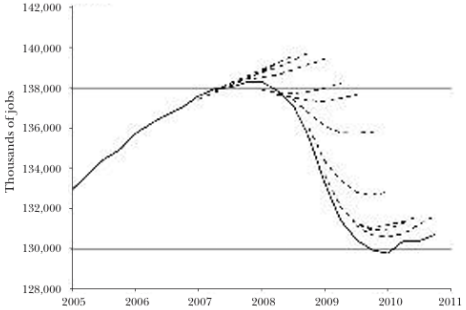
6 DSGE Estimation

7 Inference

8 Conclusion

*Figure 10*

**"The Mother of All Forecast Errors": Survey of Professional Forecasters Median Forecast for Nonfarm Business Employment during the 2007–2009 Recession and Early Recovery**



## Challenges Ahead - Big Data

- Better forecasting models to capture the severity of shocks;
- Parametrics restrictions (or priors) that make DFM and BVAR discard potentially important information;
- DFM and BVAR be improved by revealing non linear structures with machile learning tools;

## Challenges Ahead - Low Frequency Data

- Why does historical evidence tell us about whether the recent slowdown in US productivity is permanent or temporary?
- How to better identify comovoments in low-frequency data without using a particular model.
- Identification and estimation of macro models by using microeconometric data.

## And in the last 10 years?

- Local projections;
- Identification using micro and regional data;
- ML for forecasting;
- HANK models;
- Covid and 2021-2023 inflation posed new challenges to fitting models to the data.

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